

CLEVE, Australia

WMO: 946620

Lat: 33.7081S

Lon: 136.5025E

Elev: 176

StdP: 99.23

Time Zone: 9.50 (AUA)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.6	5.5	-6.4	2.2	24.6	-4.1	2.7	24.7	14.5	13.1	13.5	13.6	3.2	20	0.600

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.5	38.2	18.2	35.5	17.5	32.9	17.0	21.1	28.4	20.1	28.3	19.2	27.3	7.5	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	14.5	22.5	18.0	13.2	21.8	16.8	12.3	21.0	62.1	27.8	58.5	28.2	55.5	27.6	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.7	11.3	10.2	DB	2.7	42.7	0.7	1.6	2.2	43.9	1.8	44.8	1.4	45.6	0.9	46.8	(4)
(5)			WB	2.1	22.7	0.6	1.3	1.7	23.6	1.4	24.3	1.0	25.0	0.6	26.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.5	22.0	21.7	19.8	17.6	14.5	11.9	11.3	11.8	13.9	15.9	18.4	20.0	(6)
(7)		DBStd	5.19	4.38	4.08	3.97	3.53	2.96	2.17	2.21	2.72	3.50	4.19	4.43	4.29	(7)
(8)		HDD10.0	29	0	0	0	0	1	6	11	9	2	1	0	0	(8)
(9)		HDD18.3	1158	7	7	25	55	127	193	218	205	141	102	52	26	(9)
(10)		CDD10.0	2414	371	327	303	228	139	63	51	64	120	184	253	311	(10)
(11)		CDD18.3	503	119	101	69	33	8	1	0	1	9	27	55	79	(11)
(12)		CDH23.3	5804	1416	1046	722	306	34	1	0	7	106	397	743	1027	(12)
(13)	CDH26.7	2889	798	549	343	104	4	0	0	1	25	158	365	543	(13)	
(14)	Wind	WSAvg	5.4	5.8	5.6	5.2	4.9	5.0	5.1	5.6	5.4	5.6	5.5	5.7	(14)	
(15)	Precipitation	PrecAvg	402	18	22	22	28	43	46	52	49	43	34	26	25	(15)
(16)		PrecMax	597	76	125	91	99	118	122	101	97	120	96	78	89	(16)
(17)		PrecMin	227	1	0	0	1	5	10	14	6	12	2	0	2	(17)
(18)		PrecStd	94	18	27	21	23	25	24	22	21	26	24	18	20	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.0	40.1	37.9	33.7	26.8	21.0	20.2	24.3	29.8	34.6	38.1	40.1	(19)
(20)			MCWB	18.7	19.3	17.5	16.7	14.5	12.8	11.6	12.5	13.9	15.5	17.5	18.1	(20)
(21)		2%	DB	38.3	36.5	33.9	29.5	23.8	18.4	18.0	20.9	26.4	31.0	34.3	36.4	(21)
(22)			MCWB	18.4	18.3	17.1	15.4	13.4	11.5	10.9	11.6	13.5	14.5	16.3	17.2	(22)
(23)		5%	DB	34.7	32.9	30.7	26.6	21.3	16.8	16.4	18.5	23.7	27.8	31.2	32.6	(23)
(24)			MCWB	18.3	18.2	16.8	14.6	12.6	11.2	10.7	10.9	12.9	13.9	15.7	16.6	(24)
(25)		10%	DB	31.1	29.8	27.2	24.3	19.4	15.5	15.0	16.7	21.2	24.8	27.6	29.1	(25)
(26)			MCWB	17.5	17.9	16.3	14.1	12.6	11.1	10.4	10.5	12.1	13.2	15.1	16.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.2	23.0	21.6	18.8	16.3	14.5	13.6	14.0	16.4	17.6	19.5	20.6	(27)
(28)			MCDWB	30.4	28.6	27.1	23.7	21.0	17.5	16.2	19.5	23.5	26.5	28.7	28.8	(28)
(29)		2%	WB	20.8	21.4	20.0	17.5	15.1	13.2	12.5	12.9	14.9	16.1	18.1	19.4	(29)
(30)			MCDWB	30.4	28.1	25.9	22.8	19.5	16.0	15.4	17.6	23.0	25.4	28.5	28.6	(30)
(31)		5%	WB	19.8	20.2	18.8	16.5	14.2	12.5	11.8	12.1	13.8	15.1	17.2	18.3	(31)
(32)			MCDWB	29.9	27.3	25.1	22.2	18.5	15.1	14.7	16.6	20.8	23.8	27.1	27.4	(32)
(33)		10%	WB	18.8	19.3	17.7	15.7	13.5	11.8	11.1	11.3	12.9	14.2	16.1	17.3	(33)
(34)			MCDWB	27.7	26.6	24.3	21.5	17.7	14.3	14.0	15.3	18.8	22.1	24.4	25.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.5	10.6	10.1	9.8	7.7	6.7	6.9	8.4	10.5	12.1	12.1	12.0	(35)
(36)			MCDBR	18.8	17.4	15.9	13.9	10.5	8.3	8.3	11.2	15.0	18.0	18.3	19.1	(36)
(37)			MCWBR	5.1	4.8	4.8	4.7	4.3	4.3	4.4	5.1	5.7	6.1	5.4	5.5	(37)
(38)		5% WB	MCDBR	15.0	12.6	11.8	10.9	8.2	6.7	7.1	9.1	12.7	14.7	15.1	14.4	(38)
(39)			MCWBR	5.1	4.7	4.7	4.6	4.3	4.3	4.5	5.1	5.7	6.0	5.5	5.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.341	0.339	0.320	0.314	0.304	0.293	0.291	0.299	0.325	0.336	0.339	0.332	(40)	
(41)		taud	2.518	2.546	2.583	2.581	2.586	2.600	2.601	2.566	2.487	2.458	2.486	2.528	(41)	
(42)		Ebn at Noon	996	978	962	913	869	854	876	916	941	969	992	1008	(42)	
(43)		Edh at Noon	112	105	94	84	74	68	72	84	102	113	115	112	(43)	
(44)	All-Sky Solar Radiation	RadAvg	7.79	6.85	5.61	4.15	2.95	2.48	2.71	3.62	4.89	6.19	7.14	7.62	(44)	
(45)		RadStd	0.40	0.41	0.28	0.25	0.12	0.13	0.11	0.22	0.34	0.42	0.37	0.38	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (18 neighbors)	+0.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+104	N/A

Nomenclature: See separate page